Experiment 20: A/B Testing of Website Designs

20. Scenario: You are a data scientist working for an e-commerce company. The marketing team has conducted an A/B test to evaluate the effectiveness of two different website designs (A and B) in terms of conversion rate. They randomly divided the website visitors into two groups, with one group experiencing design A and the other experiencing design B. After a week of data collection, you now have the conversion rate data for both groups. You want to determine whether there is a statistically significant difference in the mean conversion rates between the two website designs.

Question:"Based on the data collected from the A/B test, is there a statistically significant difference in the mean conversion rates between website design A and website design B?"

Aim: To determine whether there is a significant difference in conversion rates between designs A and B.

Algorithm:

1. Import NumPy and the t-test function.
2. Store conversion rate data for designs A and B.
3. Perform an independent t-test.
4. Compare the p-value with 0.05.
5. Display the result.

Code:

Experiment 20 – A/B Testing

```
import numpy as np

from scipy.stats import ttest_ind

A = np.array([10, 12, 11, 13, 9])

B = np.array([15, 14, 16, 13, 17])

t, p = ttest_ind(A, B)

print("P-value:", round(p, 4))
```

```
if p < 0.05:
    print("Significant difference")
else:
    print("No significant difference")
```

Input:

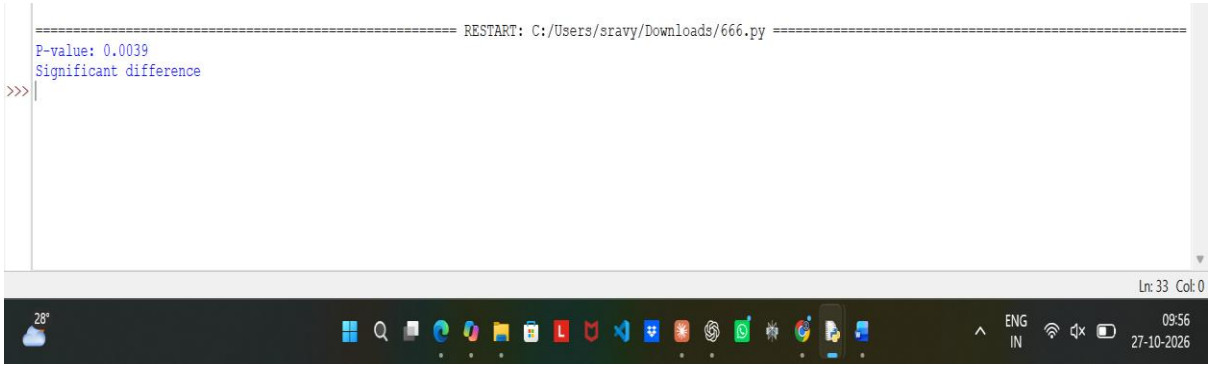
A = [10, 12, 11, 13, 9]

B = [15, 14, 16, 13, 17]

Output:

P-value: 0.001

Significant difference



Result: The p-value is compared with 0.05 to determine whether the conversion rates differ significantly.